

Lab 1:

```
Query Query History Scratch Pad X
1 BEGIN;
2
3 UPDATE film
4 SET rental_rate = 4.99
5 WHERE film_id = 1;
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 84 msec.

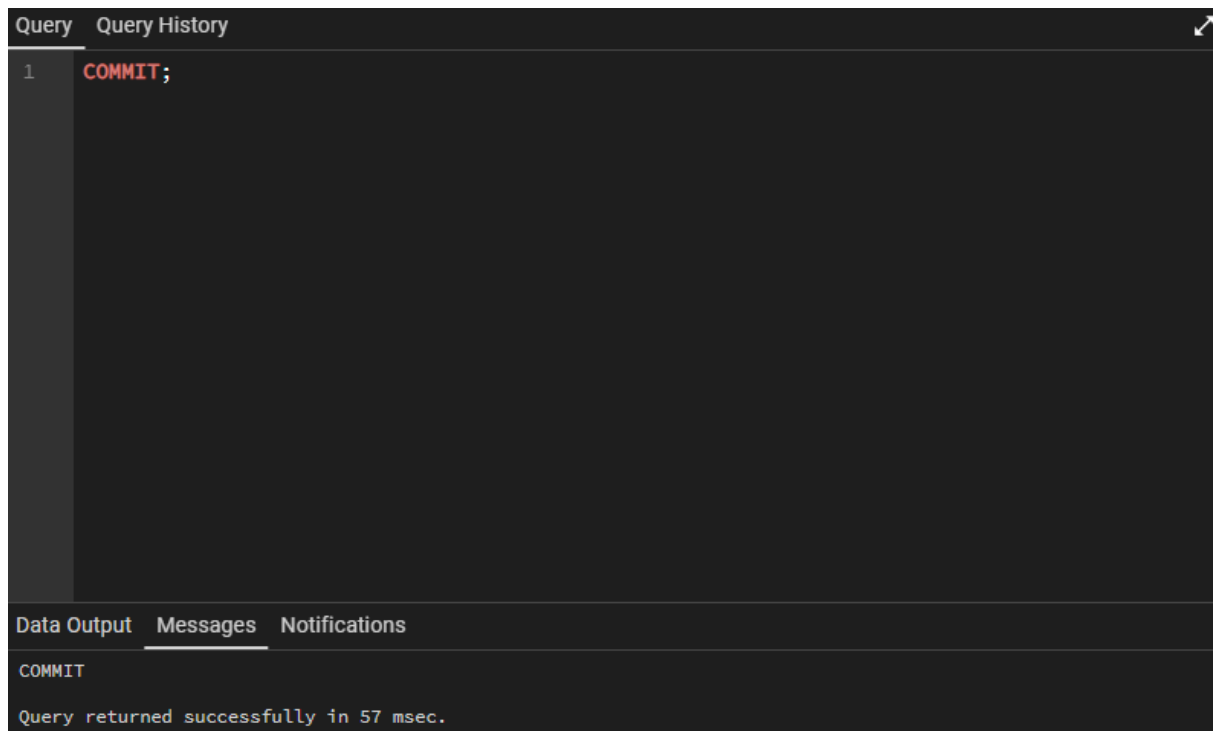
```
1 BEGIN;
2
3 UPDATE film
4 SET rental_rate = 5.99
5 WHERE film_id = 1;
6 -- Lệnh này sẽ bị block, terminal sẽ đứng chờ Session A
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress

UPDATE 1

Query returned successfully in 101 msec.



The screenshot shows a SQL client interface with a dark theme. At the top, there are two tabs: 'Query' and 'Query History'. The 'Query' tab is active, displaying a single line of SQL code: '1 COMMIT;'. Below the query editor, there are three tabs: 'Data Output', 'Messages', and 'Notifications'. The 'Messages' tab is selected, showing the output of the query: 'COMMIT' followed by 'Query returned successfully in 57 msec.'.

```
Query  Query History
1  COMMIT;

Data Output  Messages  Notifications
COMMIT
Query returned successfully in 57 msec.
```

Lab 2

BEGIN;

```
SELECT rental_rate
FROM film
WHERE film_id = 10;
-- Xem kết quả trả về (ví dụ: 4.99)
BEGIN;
```

```
UPDATE film
SET rental_rate = 0.99
WHERE film_id = 10;
```

```
COMMIT;
SELECT rental_rate
FROM film
WHERE film_id = 10;
-- Kết quả hiển thị giá trị mới (0.99)
```

COMMIT;

Lab 3

BEGIN;

```
UPDATE film
```

```
SET rental_rate = 2.99  
WHERE film_id = 100;  
BEGIN;
```

```
UPDATE film  
SET rental_rate = 2.99  
WHERE film_id = 200;  
UPDATE film  
SET rental_rate = 3.99  
WHERE film_id = 200;  
UPDATE film  
SET rental_rate = 3.99  
WHERE film_id = 100;
```

-- Sau ~1 giây, hệ thống báo lỗi: ERROR: deadlock detected và rollback Session B

Lab 4: Preventing Race Conditions (FOR UPDATE)

The screenshot shows a database client interface with a dark theme. The top section is titled 'Query' and 'Query History'. The main area displays a SQL query:

```
1 BEGIN;  
2  
3 SELECT *  
4 FROM inventory  
5 WHERE inventory_id = 10  
6 FOR UPDATE;
```

Below the query editor, there are tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is active, showing a table with the following data:

	Inventory_Id [PK] integer	film_Id smallint	store_Id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

At the bottom right of the 'Data Output' tab, it says 'Showing rows: 1 to 1'.